OPHIRA

Reflectometry-fitting toolkit for porous silicon and thin films

D3 — Operating procedures and strategies

Usage manual: step-by-step recipes and strategies to get robust results from the code (determination of *d*, KK cycle, d-scan, multi-angle, difficult cases). Complementary to **D1** (description of each control) and **D2** (physics and mathematics). The procedures here give the *how* and the *why*; for the *what a button does* see D1, for the formulas see D2.

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Related documents	D1 (user manual), D2 (physics & math)

Contents

1 How to read this manual

Each procedure is a sequence of numbered steps. Two kinds of box accompany the steps:

- **Strategy** (green): the *why* of a choice or a suggestion to improve the result.
- **Warning** (orange): a caution or a common mistake to avoid.

Prerequisite: having read D1, i.e. knowing what each button / slider / window does. Here that knowledge is assumed and the focus is on usage.

Strategy — General principle of the project: **data lead, physics validates**. The thickness d is over-determined (maxima, fit, KK consistency): the convergence of independent criteria on the same d is what makes the result solid, not a single number.

2 Procedure: new sample from scratch (8 deg)

Goal: start from a raw spectrum and reach a first reasonable fit at 8 deg, from which to refine.

1. **LOAD** the 8 deg spectrum (see D1, row 1). Only the R_{exp} are loaded: no parameters, no fit.
2. Select the **substrate** (usually Si) and **film** = PSi (free n,k sliders, free-node mode).
3. **Coarse-set the thickness**: move the *d* slider until the number and position of the simulated fringes get close to the experimental ones. Do not seek perfection, only the order of magnitude.
4. **Align n in bulk**: use **n +/**– to raise/lower the whole n(λ) curve and **slope +/**– to tilt it (λ > 460 nm), until the fringes match better.
5. **MEAS JUMP**: measures and locks the detector jump at ~890 nm from the data (instrumental anchoring, see dedicated procedure).
6. **FIT** at 8 deg. Check the fit — the **residual panel** below the spectrum and the **RMS** figure it reports; if coarse misalignments remain, adjust *d* / n +/– and rerun.

Strategy — Always start from 8 deg: KK is physically valid only at near-normal incidence and the conditioning of the problem is better. The other angles are then seeded from 8 deg (see multi-angle).

Warning — Do not chase a good **fit residual** by moving many parameters at random: first the fringe geometry (d, n in bulk), then the fit. A fit launched on completely misaligned fringes converges poorly.

3 Procedure: determination of d and KK cycle

Goal: fix the thickness *d* and refine n(λ), k(λ) to the best possible Kramers-Kronig consistency. It is the methodological core for a well-characterized sample.

1. Initial estimate of *d*: **FIT AUTO** (or 8 deg/20 deg triangulation in the Peak Analysis window, COMPUTE d (8/20) button).
2. **Lock d** (thickness slider lock): from here on *d* is fixed and you work only on n, k and instrumental parameters.
3. Single **FIT** at 8 deg with d locked.
4. Open the **KK** window (KK button at 8 deg) and press **RECALCULATE**.
5. If the disagreement RMS$(n - n_{KK})/\sigma_{KK} > 3$, press **APPLY KK→NODES** (transfers n_{KK} to the UV nodes, λ < 420–470 nm region).
6. If the 2550 nm node drifts (boundary artifact), **lock it by hand** to the value from the previous cycle.
7. Back to the main window and **FIT** again. Repeat steps 4–7 for 2–5 cycles.

Stopping criterion: RMS$(n - n_{KK})/\sigma_{KK}$ stabilized at the noise level (~2–3) with < 5% variation between successive cycles.

Strategy — Two signals that *d* is the right one: (a) the KK residual flattens to the noise floor; (b) **no jump/step in n at the KK edge (~480 nm)**. The jump at 480 nm grows and the fit residual worsens rapidly as you move away from the correct *d*: it is an objective, emergent criterion, not a by-eye judgement.

Warning — The UV region (λ < 420 nm) is data-limited and shows anomalous dispersion of PSi: APPLY KK→NODES acts there on purpose; do not force monotonicity below 420 nm (see D2, physical constraints).

Warning — The KK window's **display range** control only zooms the plots and limits what SAVE CURVES exports; it does **not** change the KK computation, which always runs on the full spectrum.

4 Procedure: semi-automatic d-SCAN (d bracket)

Goal: measure *d* as an **interval** (bracket) with its uncertainty, by comparing curated fits at different thicknesses. Useful when there is no SEM and you want a defensible error bar on *d*.

1. Choose a few *d* values around the estimate (e.g. centre ± 150 nm in 50 nm steps).
2. For each *d*: set it, coarse-align n in bulk (n +/–, slope +/–), run a **curated FIT** (even a short KK cycle), then **SAVE SESS** with a name that includes the *d*.
3. Press **d-SCAN** (row 3) and select *all* the saved sessions.
4. On the **exp vs sim reflectance** overlay, place the **ΔR band by eye** where the two curves diverge in a *d*-dependent way — wide enough to give a clear slope, kept below 880 nm (detector jump).
5. Read the **ΔR d-discriminants** (a signed corroborating output): ΔC_{IR} (IR fringe-contrast) and **ΔR** (signed sum of $R_{fit} - R_{exp}$ over the band). Each changes sign as it passes the correct *d*; its **zero-crossing brackets d**, and the green span between the two crossings is an honest bracket for *d* — a range relative to this indicator, not a rigorous uncertainty on *d*.
6. Cross-check the companion figures: the **n(λ) overlay** (the *smoothest* curve(s) are highlighted — the physical *n*(λ) follows a smooth Cauchy dispersion, so a wrong *d* shows a shoulder or an overshoot near the edge; ranked by RMS deviation from a Cauchy law over 420–1000 nm, a range when the minimum is flat, KK-free); and the **RMS-residual(d) / KK-residual window** — the RMS reflectance residual plus the KK residual, readable as a magnitude (|Δn| RMS → a minimum) or **signed** (mean($n_{fit} - n_{KK}$) → a zero-crossing) over an adjustable band — a complementary view, more operator-sensitive than ΔR.

Warning — Full automation does not work: a single fit per *d* with warm-start degrades as you move away from the starting point and produces a falsely sharp minimum. The operator's care on the individual fits is necessary: the software *analyzes*, it does not replace the fit.

Strategy — In practice the most reliable indicator of *d* is the **n(λ) overlay**: the physical dispersion follows a smooth Cauchy law, so the *d* whose n(λ) is *smoothest* — no shoulder or overshoot near the edge — is the right one. The **ΔR crossing** corroborates it: band-integrated (not peak-tracking) it is far steadier than the two-angle maxima, and read-only over the curated fits it avoids the fit's initial-condition scatter. Choose the ΔR band **per sample** — it moves with porosity; a flat or wiggly ΔR(*d*) line means that band is uninformative, so widen or reposition it. If neither the n(λ) smoothness nor any ΔR band gives a clean answer, that is an honest signal to **redo the measurement**.

Strategy — **Publication figures.** Every d-scan window has **Save PNG/PDF** plus a **W:H** field and a **Lock aspect (preview)** checkbox. Tick Lock aspect and type the *same* ratio (e.g. 16:10) on every panel: the window previews the plot-only figure at that ratio (parameter table and widgets hidden, ratio kept on resize), and Save exports exactly that — identically shaped, crisp-text panels to assemble into one composite figure (PDF = vector, no text distortion). Drag a legend off overlapping curves before saving.

5 Procedure: multi-angle fit

Goal: obtain n, k consistent across the angles, exploiting the fact that the sample's n,k are the same and only the instrumental and geometric effects change.

1. Fit **8 deg** well (previous procedures).
2. Go to the next angle and press **COPY 8 deg**: copies the 8 deg n,k (ext_override) as a seed.
3. For each angle (20/40/60): **MEAS JUMP** for that angle, adjust the angular spread if HOBL, then **FIT**.
4. **FIT MULTI** with *d* fixed: global fit with consistency constraints (weights 8 deg most, decreasing toward 60 deg), followed by per-angle refinement.

Warning — 40 deg and 60 deg are in the **HOBL** configuration (angular spread, stronger regularization, very strong optical-thickness anchor): they run in a separate thread and are slower. See D2.

Strategy — At 60 deg, treat the index you read from the fringe positions as an *effective* index n_{eff} only, not as the true n. At such oblique incidence the fringe maxima no longer separate the real index *n* from the absorption *k* — the two mix — so do not try to extract *k* at 60 deg. (The FIT/MAXIMA view, D1, shows this with an explicit disclaimer.)

Warning — If the lowest-order far-IR fringe at 60 deg shows a flat / trapezoidal top, it is an artifact of the symmetric fringe-contrast envelope (the **Fr contrast / Fr ctr vtx / Fr ctr curv** sliders), not sample physics — amplitude only, fringe positions unaffected. To recover a rounded top, move **Fr ctr vtx** out of band (a monotonic contrast ramp) and enable **MONO n** (see D2, fringe-contrast envelope).

6 Procedure: using the FIT / MAXIMA view

Goal: read n(λ) derived from the *maxima positions* (n_{eff}), as a complement or comparison to the fit.

1. First compute peaks / orders m / d in the **Peak Analysis** window (otherwise the toggle tries to compute on the fly or warns).
2. On the right panel, press **View: MAXIMA** for the current angle.
3. Read $n_{eff}(\lambda)$ at the maxima; the residual is zeroed and a per-angle disclaimer appears.
4. Return to **View: FIT** for the normal presentation.

Strategy — At 60 deg the view shows only n_{eff} (k hidden) with a strong disclaimer: it is the honest choice, because at oblique incidence the maxima do not separate n and k. At the other angles the disclaimer is milder.

Strategy — In the **Peak Analysis** window the maxima source is explicit: the **FIT / EXP** selector reads the peaks from the fitted curve or from the experimental data; switching it re-extracts the peaks and recomputes *both d* and $n_{eff}(\lambda)$. The active source is shown in the plot title.

Warning — To reconstruct n(λ) from the maxima at a thickness you choose (e.g. the d_{eff} bracketed by the d-SCAN): in Peak Analysis pick the **custom d** radio, type the value, and press **CALC n(λ)**. Useful to compare n from the maxima and n from the fit at the *same* d.

7 Procedure: anchoring the IR jump (MEAS JUMP)

Goal: fix the instrumental detector jump at ~890 nm *before* fitting, so the fit does not confuse it with n, k or scattering.

1. Load the angle's data.
2. Press **MEAS JUMP**: measures the jump from the data (ratio of the levels above/below 890 nm), sets it on the IR Jump slider and **locks** it.
3. Proceed with the fit: the jump parameter is now anchored to the data.

Strategy — The jump varies by sample and by angle (typically decreases from 8 deg to 60 deg): measure it for each angle, do not reuse an old value.

Warning — MEAS JUMP requires sufficient data around 890 nm; if missing, the button warns and changes nothing.

8 Procedure: thin sample (THIN FILM)

Goal: handle films with $d \leq 700$ nm, where the fringes are few and wide and the two-angle triangulation is fragile.

1. Enable the **THIN FILM** toggle (row 2).
2. Effects: accepted λ threshold lowered from 1000 to 600 nm; accepts even a single NIR peak ($m = 1$ assumed); enables the single-angle formula, more stable on thin samples.
3. Proceed with the fit at 8 deg as usual.

Warning — On thin samples a single IR peak can determine *d*: make sure that peak is not discarded (the adaptive detection preserves it, but verify that it is marked in the Peak Analysis window).

9 Procedure: save, share, reload

1. **SAVE SESS** (.psi_sess): the whole session (data, parameters, ext_override, kk_cache). For internal backup and **exact resume**.
2. **SAVE ALL**: .par + _data.csv files for each angle, plus a manifest *name*_all.csv which is the single entry point to reload everything.
3. **LOAD**: auto-detect — if you select the manifest it loads all the angles; otherwise a single spectrum.

Strategy — Before an experiment (e.g. a d-scan or an aggressive round of KK cycles) do a **SAVE SESS**: if the exploration worsens the fit, you reload the exact state and restart.

10 Troubleshooting

10.1 The fit residual does not go down

- Check that **MEAS JUMP** has been done for that angle: a wrong IR jump left free confuses the fit.
- Check that *d* is in the right neighbourhood (coarse-set it with the slider and look at the fringe alignment) before insisting with the fit.
- Make sure you do not have too many free parameters at once: lock what is already good (slider locks).

10.2 Poor fit at ONE angle only (the others are fine)

First suspect: **wrong file loaded in that slot**. The program does not verify that the spectrum matches the slot's angle (see D1) → a file placed in the wrong angle is fitted with *that* angle's model, giving a very high residual only there, with no warning. Check the file name loaded in the slot; watch out for similar names (e.g. …`_40°` / …`_60°`).

10.3 Forgotten locked nodes

It happens often: a node or an *n/k* slider left **locked** from a previous session does not move in the fit, and the fit seems to "fail" to improve in that spectral region for no apparent reason. Before concluding that the fit is bad, **check the locks** (sliders and nodes): unlock them if you did not want them fixed and rerun.

10.4 Step in n at the KK edge (~480 nm)

It is the typical symptom of a wrong *d*: the step grows as you move away from the correct *d*. Use the d-scan to find the *d* that flattens it.

10.5 Physically absurd n in the infrared

If after an exploration n in the NIR drops to non-physical values (e.g. < 1.1), the state is degraded: **reload the last good session** (SAVE SESS) instead of trying to patch over it.

10.6 When to redo the measurement

A fit that only *looks* weak is not a reason to redo the measurement: *n* and *k* can be perfectly well defined even when the residual is large, because they are fixed by the fringe **positions** and the dispersion, not by the fringe amplitude/contrast (which scattering and inhomogeneity lower without touching n, k). Redo the measurement only when *no d* matches the fringe positions and the KK residual stays high everywhere — a genuine limit in the data (non-ideal sample, inhomogeneity, alignment), not merely a low-contrast fit.

10.7 2550 nm node fitted too low

The far-IR node (~2550 nm) is **often** fitted lower than the adjacent nodes. It must be **checked**, especially before the KK: in that region the Hilbert integral is truncated (boundary artifact at 2550, see D2 and the KK window) and an anomalous 2550 distorts n_KK. Practical check: if it drifts downward, **lock it by hand** to a value consistent with the neighbouring nodes before RECALCULATE.